EduLLM Type 1 — Top 100 MCQs

Class 10 Science | Chemical Reactions and Equations

Rephrased questions with concept-recall explanations. Teacher review flags are shown where the source is ambiguous.

Q1. What best describes a chemical reaction?

A. Only physical properties change
B. New substances are produced
C. Only the shape changes
D. Only the physical state changes

Answer: B | CRE-SIGNS-01 | Easy

Recall: A chemical reaction creates one or more substances with properties different from the reactants.

Q2. Which combination of observations may indicate that a chemical reaction occurred?

A. A colour change
B. Release of a gas
C. Formation of a precipitate
D. All of these

Answer: D | CRE-SIGNS-01 | Easy

Recall: Colour change, gas evolution and precipitate formation are common evidence of chemical reactions.

Q3. What is produced when magnesium ribbon burns in air?

A. Magnesium chloride
B. Magnesium sulphate
C. Magnesium oxide
D. Magnesium nitrate

Answer: C | CRE-COMB-01 | Easy

Recall: Magnesium combines with oxygen to form white magnesium oxide: $2Mg + O_2 \rightarrow 2MgO$.

Q4. Select the balanced equation for magnesium burning in oxygen.

A. $Mg + O_2 \rightarrow MgO$
B. $2Mg + O_2 \rightarrow 2MgO$
C. $Mg + 2O_2 \rightarrow MgO$
D. $MgO + O_2 \rightarrow Mg$

Answer: B | CRE-BALANCE-01 | Moderate

Recall: Two magnesium atoms react with one oxygen molecule to produce two units of MgO.

Q5. Which principle must a balanced chemical equation obey?

A. Newton's laws of motion
B. Law of gravitation
C. Law of conservation of mass
D. Ohm's law

Answer: C | CRE-BALANCE-01 | Moderate

Recall: The number of atoms of every element is equal on both sides, conserving mass.

Q6. Where are the reactants placed in a conventional chemical equation?

A. To the right of the arrow
B. On both sides of the arrow
C. To the left of the arrow
D. Below the arrow

Answer: C | CRE-EQUATION-01 | Easy

Recall: Reactants are written before the arrow; products are written after it.

Q7. On which side of a chemical equation are products normally shown?

A. Right side
B. Left side
C. Above the arrow
D. Below the arrow

Answer: A | CRE-EQUATION-01 | Easy

Recall: Products are written on the right-hand side of the reaction arrow.

Q8. Which equation correctly balances hydrogen reacting with oxygen to make water?

A. $H_2 + O_2 \rightarrow H_2O$
B. $H_2 + O_2 \rightarrow 2H_2O$
C. $2H_2 + O_2 \rightarrow 2H_2O$
D. $H_2 + 2O_2 \rightarrow H_2O$

Answer: C | CRE-BALANCE-01 | Moderate

Recall: $2H_2 + O_2 \rightarrow 2H_2O$ balances four hydrogen atoms and two oxygen atoms.

Q9. Which reaction illustrates two reactants combining to form one product?

A. $CaO \rightarrow Ca + O_2$
B. $2H_2 + O_2 \rightarrow 2H_2O$
C. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
D. $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: B | CRE-COMB-01 | Easy

Recall: Hydrogen and oxygen combine to make only water, so this is a combination reaction.

Q10. Which equation is an example of decomposition?

A. $NaOH + HCl \rightarrow NaCl + H_2O$
B. $CaCO_3 \rightarrow CaO + CO_2$
C. $2Na + Cl_2 \rightarrow 2NaCl$
D. $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$

Answer: B | CRE-DECOMP-01 | Moderate

Recall: One compound, $CaCO_3$, breaks into CaO and CO_2 on heating.

Q11. In which reaction does one element replace another from a compound?

A. $2H_2 + O_2 \rightarrow 2H_2O$
B. $Cu + 2AgNO_3 \rightarrow Cu(NO_3)_2 + 2Ag$
C. $2KClO_3 \rightarrow 2KCl + 3O_2$
D. $CaO + H_2O \rightarrow Ca(OH)_2$

Answer: B | CRE-DISPLACE-01 | Easy

Recall: Copper displaces silver from silver nitrate because copper is more reactive than silver.

Q12. Which reaction involves an exchange of ions between two compounds?

A. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
B. $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
C. $2Mg + O_2 \rightarrow 2MgO$
D. $CaCO_3 \rightarrow CaO + CO_2$

Answer: B | CRE-DOUBLE-01 | Moderate

Recall: Silver nitrate and sodium chloride exchange ions and form insoluble AgCl.

Q13. Identify the oxidation–reduction reaction.

A. $NaCl + AgNO_3 \rightarrow AgCl + NaNO_3$
B. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
C. $CaCO_3 \rightarrow CaO + CO_2$
D. $HCl + NaOH \rightarrow NaCl + H_2O$

Answer: B | CRE-REDOX-01 | Moderate

Recall: Zinc loses electrons and copper ions gain electrons; oxidation and reduction occur together.

Q14. What happens when an iron nail is placed in copper sulphate solution?

A. No reaction occurs
B. Iron sulphate forms and copper deposits
C. Iron oxide and copper form
D. Copper sulphate and iron form

Answer: B | CRE-DISPLACE-01 | Easy

Recall: Iron displaces copper: $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$; a brown copper layer appears.

Q15. Which gas is liberated when zinc reacts with dilute hydrochloric acid?

A. Oxygen
B. Carbon dioxide
C. Hydrogen
D. Nitrogen

Answer: C | CRE-METAL-ACID-01 | Easy

Recall: $Zn + 2HCl \rightarrow ZnCl_2 + H_2$. Hydrogen can be identified by a pop sound.

Q16. Which equation balances the reaction between iron and steam?

A. $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
B. $3Fe + 4H_2O \rightarrow Fe_2O_3 + 4H_2$
C. $Fe + 4H_2O \rightarrow Fe_3O_4 + H_2$
D. $2Fe + 3H_2O \rightarrow Fe_2O_3 + 3H_2$

Answer: A | CRE-BALANCE-01 | Moderate

Recall: Hot iron reacts with steam to form Fe_3O_4 and hydrogen: $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$.

Q17. Which process releases heat to the surroundings?

A. Photosynthesis
B. Thermal decomposition of $CaCO_3$
C. Burning of coal
D. Electrolysis of water

Answer: C | CRE-EXOTHERM-01 | Easy

Recall: Coal combustion releases heat and is therefore exothermic.

Q18. Which process requires heat to proceed?

A. Respiration

B. Digestion of food
C. Formation of slaked lime
D. Thermal decomposition of $CaCO_3$

Answer: D | CRE-ENDOTHERM-01 | Easy

Recall: Calcium carbonate absorbs heat to break into calcium oxide and carbon dioxide.

Q19. Which reaction is classified as a combination reaction?

A. $CaO + CO_2 \rightarrow CaCO_3$
B. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
C. $2Na + 2H_2O \rightarrow 2NaOH + H_2$
D. $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: A | CRE-COMB-01 | Easy

Recall: Two reactants join to form one product: $CaO + CO_2 \rightarrow CaCO_3$.

Q20. Which equation represents the decomposition of hydrogen peroxide?

A. $2H_2 + O_2 \rightarrow 2H_2O$
B. $2H_2O_2 \rightarrow 2H_2O + O_2$
C. $2Na + 2HCl \rightarrow 2NaCl + H_2$
D. $CaO + H_2O \rightarrow Ca(OH)_2$

Answer: B | CRE-DECOMP-01 | Moderate

Recall: Hydrogen peroxide decomposes into water and oxygen, often accelerated by a catalyst.

Q21. Which reaction is a single displacement process?

A. $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
B. $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
C. $2H_2 + O_2 \rightarrow 2H_2O$
D. $CaCO_3 \rightarrow CaO + CO_2$

Answer: B | CRE-DISPLACE-01 | Easy

Recall: Zinc displaces hydrogen from dilute sulphuric acid, releasing H_2 gas.

Q22. Which equation shows a double displacement reaction?

A. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B. $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$
C. $2KClO_3 \rightarrow 2KCl + 3O_2$
D. $Mg + 2HCl \rightarrow MgCl_2 + H_2$

Answer: B | CRE-DOUBLE-01 | Moderate

Recall: Barium and sodium salts exchange ions, producing insoluble $BaSO_4$.

Q23. Which equation demonstrates precipitation?

A. $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$
B. $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
C. $2H_2 + O_2 \rightarrow 2H_2O$
D. $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: A | CRE-PRECIP-01 | Easy

Recall: Barium sulphate forms as an insoluble white solid called a precipitate.

Q24. Which reaction is an acid–base neutralisation?

A. $HCl + NaOH \rightarrow NaCl + H_2O$
B. $CaCO_3 \rightarrow CaO + CO_2$

C. $2H_2 + O_2 \rightarrow 2H_2O$
D. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

Answer: A | CRE-NEUTRAL-01 | Easy

Recall: Acid and base react to form salt and water.

Q25. Which equation correctly balances iron reacting with steam?

A. $Fe + H_2O \rightarrow Fe_3O_4 + H_2$
B. $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$
C. $Fe + 2H_2O \rightarrow FeO_2 + 2H_2$
D. $2Fe + 3H_2O \rightarrow Fe_2O_3 + 3H_2$

Answer: B | CRE-BALANCE-01 | Moderate

Recall: $3Fe + 4H_2O \rightarrow Fe_3O_4 + 4H_2$ balances Fe, H and O atoms.

Q26. Which reaction is a single displacement reaction?

A. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B. $2Na + Cl_2 \rightarrow 2NaCl$
C. $CaCO_3 \rightarrow CaO + CO_2$
D. $2AgNO_3 + Cu \rightarrow Cu(NO_3)_2 + 2Ag$

Answer: A | CRE-DISPLACE-01 | Easy

Recall: Zinc replaces copper in copper sulphate solution. Note: option D is also a displacement reaction in the source, making this item ambiguous.

Teacher review required

Q27. Which reaction is NOT a decomposition reaction?

A. $2H_2O_2 \rightarrow 2H_2O + O_2$
B. $CaCO_3 \rightarrow CaO + CO_2$
C. $2KClO_3 \rightarrow 2KCl + 3O_2$
D. $H_2 + Cl_2 \rightarrow 2HCl$

Answer: D | CRE-DECOMP-01 | Moderate

Recall: Hydrogen and chlorine combine into one compound; the other reactions break compounds apart.

Q28. Which example shows two reactants forming a single product?

A. $C + O_2 \rightarrow CO_2$
B. $2HCl \rightarrow H_2 + Cl_2$
C. $Zn + H_2SO_4 \rightarrow ZnSO_4 + H_2$
D. $CaCO_3 \rightarrow CaO + CO_2$

Answer: A | CRE-COMB-01 | Easy

Recall: Carbon and oxygen combine to produce carbon dioxide.

Q29. Which of these reactions involves both oxidation and reduction?

A. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B. $Zn + NaOH \rightarrow$ products
C. $CaCO_3 \rightarrow CaO + CO_2$
D. $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: A | CRE-REDOX-01 | Moderate

Recall: Zinc is oxidised to Zn^{2+} and Cu^{2+} is reduced to copper metal.

Q30. Which reaction results in an insoluble solid forming?

A. $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

B. $HCl + NaOH \rightarrow NaCl + H_2O$
C. $2H_2 + O_2 \rightarrow 2H_2O$
D. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Answer: A | CRE-PRECIP-01 | Easy

Recall: Silver chloride is insoluble in water and appears as a white precipitate.

Q31. What is released when dilute hydrochloric acid reacts with zinc?

A. No reaction occurs
B. Hydrogen gas
C. Oxygen gas
D. Carbon dioxide

Answer: B | CRE-METAL-ACID-01 | Easy

Recall: Zinc displaces hydrogen from acid, forming $ZnCl_2$ and H_2 gas.

Q32. After an iron nail stays in copper sulphate solution, what colour change is expected in the solution?

A. Blue
B. Colourless
C. Green
D. Yellow

Answer: B | CRE-DISPLACE-01 | Easy

Recall: The source marks colourless, but the expected result is pale green because $FeSO_4$ forms. This source item must be corrected before students attempt it.

Teacher review required

Q33. Which gas is produced when marble chips react with dilute hydrochloric acid?

A. Hydrogen
B. Oxygen
C. Carbon dioxide
D. Nitrogen

Answer: C | CRE-CARBONATE-01 | Easy

Recall: $CaCO_3 + 2HCl \rightarrow CaCl_2 + H_2O + CO_2$.

Q34. Which equation correctly shows sodium reacting with water?

A. $Na + H_2O \rightarrow NaOH + H_2$
B. $2Na + 2H_2O \rightarrow 2NaOH + H_2$
C. $2Na + H_2O \rightarrow NaOH + 2H_2$
D. $Na + 2H_2O \rightarrow 2NaOH + H_2$

Answer: B | CRE-BALANCE-01 | Moderate

Recall: Two sodium atoms react with two water molecules to form two NaOH and one H_2.

Q35. How should the breakdown of calcium carbonate on heating be classified?

A. Combination reaction
B. Decomposition reaction
C. Displacement reaction
D. Double displacement reaction

Answer: B | CRE-DECOMP-01 | Moderate

Recall: $CaCO_3 \rightarrow CaO + CO_2$: one compound splits into two substances.

Q36. Which reaction is an example of redox?

A. $BaCl_2 + Na_2SO_4 \rightarrow BaSO_4 + 2NaCl$
B. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
C. $HCl + NaOH \rightarrow NaCl + H_2O$
D. $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: B | CRE-REDOX-01 | Moderate

Recall: Zinc loses electrons while copper ions gain electrons.

Q37. Which reaction is NOT a combination reaction?

A. $2Mg + O_2 \rightarrow 2MgO$
B. $CaO + H_2O \rightarrow Ca(OH)_2$
C. $2Na + Cl_2 \rightarrow 2NaCl$
D. $CaCO_3 \rightarrow CaO + CO_2$

Answer: D | CRE-COMB-01 | Easy

Recall: $CaCO_3$ breaks down rather than combining with another substance.

Q38. Why do we balance chemical equations?

A. To improve their appearance
B. To obey conservation of mass
C. To increase reaction speed
D. To change the physical state

Answer: B | CRE-BALANCE-01 | Moderate

Recall: Equal numbers of each type of atom must appear on both sides of an equation.

Q39. Which reaction represents displacement?

A. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
B. $2Na + Cl_2 \rightarrow 2NaCl$
C. $CaCO_3 \rightarrow CaO + CO_2$
D. $2H_2 + O_2 \rightarrow 2H_2O$

Answer: A | CRE-DISPLACE-01 | Easy

Recall: Zinc displaces hydrogen from hydrochloric acid.

Q40. Which gas makes limewater appear milky?

A. Oxygen
B. Carbon dioxide
C. Nitrogen
D. Hydrogen

Answer: B | CRE-CO2-TEST-01 | Easy

Recall: CO_2 reacts with limewater to produce insoluble $CaCO_3$, causing milkiness.

Q41. Which reaction is an example of neutralisation?

A. $H_2SO_4 + 2NaOH \rightarrow Na_2SO_4 + 2H_2O$
B. $2KClO_3 \rightarrow 2KCl + 3O_2$
C. $2H_2 + O_2 \rightarrow 2H_2O$
D. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$

Answer: A | CRE-NEUTRAL-01 | Easy

Recall: Sulphuric acid and sodium hydroxide form a salt and water.

Q42. Which equation correctly balances methane combustion?

A. $CH_4 + O_2 \rightarrow CO_2 + H_2O$
B. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
C. $CH_4 + O_2 \rightarrow 2CO_2 + 2H_2O$
D. $2CH_4 + O_2 \rightarrow 2CO_2 + H_2O$

Answer: B | CRE-BALANCE-01 | Moderate

Recall: $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$ balances carbon, hydrogen and oxygen.

Q43. When aluminium reacts with dilute hydrochloric acid, which gas forms?

A. Oxygen
B. Hydrogen
C. Nitrogen
D. Carbon dioxide

Answer: B | CRE-METAL-ACID-01 | Easy

Recall: $2Al + 6HCl \rightarrow 2AlCl_3 + 3H_2$.

Q44. Which reaction includes electron transfer?

A. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
B. $CaCO_3 \rightarrow CaO + CO_2$
C. $HCl + NaOH \rightarrow NaCl + H_2O$
D. $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: A | CRE-REDOX-01 | Moderate

Recall: Zinc is oxidised and hydrogen ions are reduced to hydrogen gas.

Q45. What forms when potassium hydroxide is added to iron(II) sulphate solution?

A. $Fe(OH)_2$
B. $Fe(OH)_3$
C. Fe_2O_3
D. FeO

Answer: A | CRE-PRECIP-01 | Easy

Recall: $FeSO_4 + 2KOH \rightarrow Fe(OH)_2\downarrow + K_2SO_4$; iron(II) hydroxide is a green precipitate.

Q46. Which process is an example of oxidation?

A. Addition of oxygen
B. Addition of hydrogen
C. Removal of oxygen
D. Removal of hydrogen only

Answer: A | CRE-OXIDATION-01 | Easy

Recall: Oxidation can mean addition of oxygen or removal of hydrogen (and, more generally, electron loss).

Q47. What black compound forms when copper is heated in air?

A. CuO
B. Cu_2O
C. $CuSO_4$
D. $Cu(OH)_2$

Answer: A | CRE-OXIDATION-01 | Easy

Recall: Heated copper reacts with oxygen to form black copper(II) oxide: $2Cu + O_2 \rightarrow 2CuO$.

Q48. Which equation represents a combination reaction?

A. $2SO_2 + O_2 \rightarrow 2SO_3$
B. $CaCO_3 \rightarrow CaO + CO_2$
C. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
D. $CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

Answer: A | CRE-COMB-01 | Easy

Recall: Sulphur dioxide and oxygen form a single product, sulphur trioxide.

Q49. What is the chemical formula of baking soda?

A. Na_2CO_3
B. $NaHCO_3$
C. NaOH
D. Na_2SO_4

Answer: B | CRE-DAILY-01 | Easy

Recall: Baking soda is sodium hydrogen carbonate, $NaHCO_3$.

Q50. In a balanced chemical equation, where are atom counts equal?

A. Only on the left side
B. Only on the right side
C. On both sides
D. Neither side

Answer: C | CRE-BALANCE-01 | Moderate

Recall: A balanced equation has the same number of atoms of each element among reactants and products.

Q51. Which reaction shows potassium chlorate decomposing?

A. $2KClO_3 \rightarrow 2KCl + 3O_2$
B. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
C. $CaCO_3 \rightarrow CaO + CO_2$
D. $2Na + Cl_2 \rightarrow 2NaCl$

Answer: A | CRE-DECOMP-01 | Moderate

Recall: Potassium chlorate releases oxygen on heating, often with MnO_2 as catalyst.

Q52. How many different products are formed in a typical combination reaction?

A. Two products
B. One product
C. No product
D. Many products

Answer: B | CRE-COMB-01 | Easy

Recall: Two or more reactants combine to form a single product.

Q53. Which reaction shows a double displacement process?

A. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B. $2H_2 + O_2 \rightarrow 2H_2O$
C. $2KClO_3 \rightarrow 2KCl + 3O_2$
D. $O_2 + C \rightarrow CO_2$

Answer: A | CRE-DOUBLE-01 | Moderate

Recall: Source key says A, but $Zn + CuSO_4$ is single displacement; no listed option represents double displacement.

Teacher review required

Q54. How is the general equation A + BC → AC + B classified?

A. Combination reaction
B. Decomposition reaction
C. Single displacement reaction
D. Double displacement reaction

Answer: C | CRE-DISPLACE-01 | Easy

Recall: A replaces B in BC to form AC and free B.

Q55. Which reaction represents single displacement?

A. $Na_2SO_4 + BaCl_2 \rightarrow BaSO_4 + 2NaCl$
B. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
C. $2H_2 + O_2 \rightarrow 2H_2O$
D. $CaCO_3 \rightarrow CaO + CO_2$

Answer: B | CRE-DISPLACE-01 | Easy

Recall: Zinc displaces hydrogen from HCl.

Q56. Which gas is released when dilute HCl reacts with a metal carbonate?

A. Oxygen
B. Hydrogen
C. Carbon dioxide
D. Nitrogen

Answer: C | CRE-CARBONATE-01 | Easy

Recall: Acid + carbonate → salt + water + CO_2.

Q57. Which reaction represents neutralisation?

A. $HCl + NaOH \rightarrow NaCl + H_2O$
B. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
C. $CaCO_3 \rightarrow CaO + CO_2$
D. $2H_2 + O_2 \rightarrow 2H_2O$

Answer: A | CRE-NEUTRAL-01 | Easy

Recall: An acid and a base produce salt and water.

Q58. How is calcium carbonate breaking down on heating classified?

A. Combination
B. Decomposition
C. Displacement
D. Redox

Answer: B | CRE-DECOMP-01 | Moderate

Recall: Heat splits $CaCO_3$ into CaO and CO_2.

Q59. Which reaction is a redox reaction?

A. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B. $2H_2 + O_2 \rightarrow 2H_2O$
C. $CaCO_3 \rightarrow CaO + CO_2$
D. $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

Answer: A | CRE-REDOX-01 | Moderate

Recall: Zinc is oxidised and copper ions are reduced. Hydrogen combustion in option B is also redox, so this source item is ambiguous.

Teacher review required

Q60. Which equation illustrates decomposition in preparing oxygen?

A. $2H_2O_2 \rightarrow 2H_2O + O_2$
B. $2KClO_3 \rightarrow 2KCl + 3O_2$
C. $2H_2 + O_2 \rightarrow 2H_2O$
D. $CaCO_3 \rightarrow CaO + CO_2$

Answer: B | CRE-DECOMP-01 | Moderate

Recall: Heating potassium chlorate produces oxygen gas.

Q61. Which reaction is an example of single displacement?

A. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B. $2H_2 + O_2 \rightarrow 2H_2O$
C. $CaCO_3 \rightarrow CaO + CO_2$
D. $2Na + Cl_2 \rightarrow 2NaCl$

Answer: A | CRE-DISPLACE-01 | Easy

Recall: Zinc displaces copper from copper sulphate.

Q62. Which substance forms as a precipitate when silver nitrate and sodium chloride react?

A. $NaNO_3$
B. $AgCl$
C. $NaCl$
D. $AgNO_3$

Answer: B | CRE-PRECIP-01 | Easy

Recall: AgCl is insoluble and forms a white precipitate.

Q63. Which observation is NOT necessarily evidence of a chemical reaction?

A. Change in colour
B. Change in physical state
C. Evolution of gas
D. Formation of a precipitate

Answer: B | CRE-SIGNS-01 | Easy

Recall: A change of state can be purely physical, such as melting ice; it is not by itself proof of a chemical reaction.

Q64. Why is a balanced chemical equation necessary?

A. It displays a colour change
B. It indicates a temperature change
C. It satisfies conservation of mass
D. It makes reactions faster

Answer: C | CRE-BALANCE-01 | Moderate

Recall: Balancing preserves the number of atoms of each element.

Q65. What forms when quicklime is mixed with water?

A. $CaCO_3$
B. $Ca(OH)_2$
C. CaO
D. CO_2

Answer: B | CRE-COMB-01 | Easy

Recall: $CaO + H_2O \rightarrow Ca(OH)_2$, called slaked lime, and heat is released.

Q66. Which reaction is represented by A + BC → AC + B?

A. Combination reaction
B. Decomposition reaction
C. Displacement reaction
D. Double displacement reaction

Answer: C | CRE-DISPLACE-01 | Easy

Recall: One element replaces another in a compound, which defines single displacement.

Q67. Which reaction represents a combination process?

A. $CaO + H_2O \rightarrow Ca(OH)_2$
B. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
C. $2AgBr \rightarrow 2Ag + Br_2$
D. $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

Answer: A | CRE-COMB-01 | Easy

Recall: Quicklime combines with water to form a single product, slaked lime.

Q68. Which equation shows decomposition?

A. $2H_2O \rightarrow 2H_2 + O_2$
B. $CaCO_3 \rightarrow CaO + CO_2$
C. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
D. $N_2 + 3H_2 \rightarrow 2NH_3$

Answer: A | CRE-DECOMP-01 | Moderate

Recall: Electrolysis breaks water into hydrogen and oxygen. Note that option B also depicts decomposition.

Teacher review required

Q69. Which equation demonstrates displacement?

A. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B. $2Na + Cl_2 \rightarrow 2NaCl$
C. $CaO + H_2O \rightarrow Ca(OH)_2$
D. $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: A | CRE-DISPLACE-01 | Easy

Recall: More reactive zinc replaces copper from its salt solution.

Q70. Which equation illustrates double displacement?

A. $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
B. $2H_2 + O_2 \rightarrow 2H_2O$
C. $CaO + H_2O \rightarrow Ca(OH)_2$
D. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Answer: A | CRE-DOUBLE-01 | Moderate

Recall: Two soluble salts exchange ions to give insoluble AgCl.

Q71. Which reaction is NOT an oxidation–reduction reaction?

A. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
B. $2H_2 + O_2 \rightarrow 2H_2O$
C. $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
D. $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

Answer: C | CRE-REDOX-01 | Moderate

Recall: Ionic exchange forming AgCl does not change oxidation states.

Teacher review required

Q72. In which reaction does hydrogen react with oxygen to produce water?

A. $2H_2 + O_2 \rightarrow 2H_2O$
B. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
C. $SO_2 + 2H_2S \rightarrow 3S + 2H_2O$
D. $2Na + 2H_2O \rightarrow 2NaOH + H_2$

Answer: A | CRE-COMB-01 | Easy

Recall: Hydrogen is oxidised by oxygen, producing water.

Q73. Which type of reaction always includes a change in oxidation state?

A. Combination reaction
B. Displacement reaction
C. Redox reaction
D. Neutralisation reaction

Answer: C | CRE-REDOX-01 | Moderate

Recall: Redox reactions involve simultaneous oxidation and reduction, reflected in oxidation-state changes.

Q74. What term describes a reaction that gives out heat?

A. Endothermic reaction
B. Exothermic reaction
C. Oxidation reaction
D. Decomposition reaction

Answer: B | CRE-EXOTHERM-01 | Easy

Recall: Exothermic reactions release heat to the surroundings.

Q75. Which reaction illustrates reduction in the extraction of iron?

A. $2H_2 + O_2 \rightarrow 2H_2O$
B. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
C. $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
D. $CaCO_3 \rightarrow CaO + CO_2$

Answer: C | CRE-REDOX-01 | Moderate

Recall: Carbon monoxide removes oxygen from iron(III) oxide to produce iron metal.

Q76. Which reaction represents neutralisation?

A. $HCl + NaOH \rightarrow NaCl + H_2O$
B. $CaCO_3 \rightarrow CaO + CO_2$
C. $2H_2 + O_2 \rightarrow 2H_2O$
D. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Answer: A | CRE-NEUTRAL-01 | Easy

Recall: Hydrochloric acid reacts with sodium hydroxide to give sodium chloride and water.

Q77. Which gas gives a characteristic pop when tested with a burning splint?

A. Oxygen
B. Hydrogen
C. Carbon dioxide
D. Nitrogen

Answer: B | CRE-GAS-TEST-01 | Easy

Recall: Hydrogen burns rapidly with a squeaky pop sound.

Q78. Which reaction is an example of displacement?

A. $CaCO_3 \rightarrow CaO + CO_2$
B. $2KClO_3 \rightarrow 2KCl + 3O_2$
C. $2H_2 + O_2 \rightarrow 2H_2O$
D. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$

Answer: D | CRE-DISPLACE-01 | Easy

Recall: Zinc replaces hydrogen in dilute hydrochloric acid.

Q79. What are the main products when methane burns completely in air?

A. CO_2 and H_2O
B. CO and H_2
C. HCl and H_2
D. SO_2 and H_2O

Answer: A | CRE-COMBUSTION-01 | Easy

Recall: Complete methane combustion gives carbon dioxide and water.

Q80. Which reaction involves oxidation?

A. $2H_2 + O_2 \rightarrow 2H_2O$
B. $CuO + H_2 \rightarrow Cu + H_2O$
C. $2Mg + O_2 \rightarrow 2MgO$
D. $CaCO_3 \rightarrow CaO + CO_2$

Answer: C | CRE-OXIDATION-01 | Easy

Recall: Magnesium gains oxygen to form magnesium oxide; option A also involves oxidation of hydrogen.

Teacher review required

Q81. What happens when quicklime is added to water?

A. Heat is absorbed
B. Heat is released
C. It turns into a gas
D. It remains unchanged

Answer: B | CRE-EXOTHERM-01 | Easy

Recall: CaO reacts vigorously with water to form $Ca(OH)_2$ and releases heat.

Q82. Which equation represents precipitation?

A. $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$
B. $2H_2 + O_2 \rightarrow 2H_2O$
C. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
D. $CaCO_3 \rightarrow CaO + CO_2$

Answer: A | CRE-PRECIP-01 | Easy

Recall: Insoluble white silver chloride forms when the solutions are mixed.

Q83. Which process releases heat?

A. Photosynthesis
B. Respiration
C. Dissolving ammonium nitrate in water
D. Electrolysis of water

Answer: B | CRE-EXOTHERM-01 | Easy

Recall: Respiration oxidises glucose and releases energy.

Q84. Which change is primarily physical rather than chemical?

A. Rusting of iron
B. Burning of paper
C. Melting of ice
D. Formation of curd

Answer: C | CRE-PHYSICAL-01 | Easy

Recall: Melting changes only the physical state; the substance remains H_2O.

Q85. Which equation is an oxidation–reduction reaction?

A. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
B. $CaCO_3 \rightarrow CaO + CO_2$
C. $2KClO_3 \rightarrow 2KCl + 3O_2$
D. $2H_2 + O_2 \rightarrow 2H_2O$

Answer: A | CRE-REDOX-01 | Moderate

Recall: Zinc loses electrons while H^+ gains electrons. Other listed reactions can also be redox, so the source question is ambiguous.

Teacher review required

Q86. Which products form when steam reacts with hot iron?

A. Iron(II) oxide and hydrogen
B. Iron(II) oxide and oxygen
C. Iron(III) oxide and hydrogen
D. Iron oxide (Fe_3O_4) and hydrogen

Answer: D | CRE-METAL-WATER-01 | Easy

Recall: $3Fe + 4H_2O(g) \rightarrow Fe_3O_4 + 4H_2$; the oxide is magnetite (mixed-valence iron oxide).

Teacher review required

Q87. Which reaction is a displacement reaction?

A. $2SO_2 + O_2 \rightarrow 2SO_3$
B. $Zn + CuSO_4 \rightarrow ZnSO_4 + Cu$
C. $CaCO_3 \rightarrow CaO + CO_2$
D. $2KClO_3 \rightarrow 2KCl + 3O_2$

Answer: B | CRE-DISPLACE-01 | Easy

Recall: Zinc displaces copper because it is more reactive.

Q88. How is a metal reacting with an acid generally classified?

A. Combination reaction
B. Decomposition reaction
C. Displacement reaction
D. Oxidation reaction

Answer: C | CRE-DISPLACE-01 | Easy

Recall: A reactive metal typically displaces hydrogen from a dilute acid.

Q89. What is the white precipitate when barium chloride and sodium sulphate solutions are mixed?

A. $BaSO_4$
B. $NaCl$
C. $BaCl_2$
D. Na_2SO_4

Answer: A | CRE-PRECIP-01 | Easy

Recall: Ba^{2+} and SO_4^{2-} combine to form insoluble barium sulphate.

Q90. Which equation shows the breakdown of baking soda on heating?

A. $2NaHCO_3 \rightarrow Na_2CO_3 + CO_2 + H_2O$
B. $2NaOH + CO_2 \rightarrow Na_2CO_3 + H_2O$
C. $Na_2CO_3 + HCl \rightarrow NaCl + CO_2 + H_2O$
D. $CaCO_3 \rightarrow CaO + CO_2$

Answer: A | CRE-DECOMP-01 | Moderate

Recall: Heating sodium hydrogen carbonate forms sodium carbonate, carbon dioxide and water.

Q91. Which substance is used to relieve acidity in the stomach?

A. Sodium bicarbonate
B. Calcium carbonate
C. Quicklime
D. Hydrochloric acid

Answer: A | CRE-DAILY-01 | Easy

Recall: Sodium bicarbonate is a mild base that can neutralise excess stomach acid.

Q92. Which gas is released when an acid reacts with a carbonate and turns limewater milky?

A. Oxygen
B. Carbon dioxide
C. Hydrogen
D. Nitrogen

Answer: B | CRE-CO2-TEST-01 | Easy

Recall: Carbonates release CO_2 with acids; CO_2 turns limewater milky.

Q93. Which compound undergoes thermal decomposition?

A. Calcium carbonate
B. Sodium chloride
C. Potassium nitrate
D. Copper(II) sulphate

Answer: A | CRE-DECOMP-01 | Moderate

Recall: On heating, $CaCO_3$ breaks into CaO and CO_2.

Q94. Which reaction is an example of redox?

A. $Zn + 2HCl \rightarrow ZnCl_2 + H_2$
B. $2KClO_3 \rightarrow 2KCl + 3O_2$
C. $CaCO_3 \rightarrow CaO + CO_2$
D. $AgNO_3 + NaCl \rightarrow AgCl + NaNO_3$

Answer: B | CRE-REDOX-01 | Moderate

Recall: Potassium chlorate decomposition involves changes in chlorine and oxygen oxidation states; option A is also redox.

Teacher review required

Q95. Which preservative helps slow rancidity in packaged foods?

A. Sodium bicarbonate
B. Sodium metabisulphite
C. Baking soda
D. Common salt

Answer: B | CRE-RANCIDITY-01 | Easy

Recall: Preservatives such as sodium metabisulphite can slow oxidative spoilage in suitable foods; its suitability depends on the food and permitted uses.

Teacher review required

Q96. Which decomposition reaction is commonly accelerated by a catalyst?

A. Decomposition of calcium carbonate
B. Combination of hydrogen and oxygen
C. Decomposition of hydrogen peroxide
D. Reaction of zinc with dilute HCl

Answer: C | CRE-CATALYST-01 | Easy

Recall: MnO_2 can catalyse H_2O_2 decomposition into H_2O and O_2.

Q97. Which equation is balanced correctly?

A. $H_2 + O_2 \rightarrow H_2O$
B. $2H_2 + O_2 \rightarrow 2H_2O$
C. $H_2 + O_2 \rightarrow 2H_2O$
D. $2H_2 + 2O_2 \rightarrow 2H_2O$

Answer: B | CRE-BALANCE-01 | Moderate

Recall: $2H_2 + O_2 \rightarrow 2H_2O$ has equal numbers of hydrogen and oxygen atoms on both sides.

Q98. Which oxide is acidic?

A. CO_2
B. CaO
C. Na_2O
D. MgO

Answer: A | CRE-OXIDES-01 | Easy

Recall: Carbon dioxide is a non-metal oxide that forms carbonic acid in water.

Q99. Which observation alone is NOT conclusive evidence of a chemical reaction?

A. Change in colour
B. Formation of a precipitate
C. Change in shape
D. Evolution of a gas

Answer: C | CRE-SIGNS-01 | Easy

Recall: A shape change can be physical; it does not necessarily create a new substance.

Q100. What is formed when quicklime reacts with water?

A. Calcium carbonate
B. Calcium hydroxide
C. Calcium oxide
D. Calcium sulphate

Answer: B | CRE-COMB-01 | Easy

Recall: Quicklime (CaO) combines with water to produce slaked lime, $Ca(OH)_2$.